Temporal Regularized Learning: Self-Supervised Learning Local in Space and Time

Davide Wiest
Department of Computer Science, TU Darmstadt
Darmstadt, Germany
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Abstract

VICReg organizes noncontrastive representation learning around invariance, marginal variance, and cross-coordinate covariance. Its usual batch statistics and two-view computation do not directly yield a causal, neuron-local update. Temporal Regularized Learning (TEREL) converts this decomposition into a causal surrogate: adjacent observations form the positive pair, detached running moments replace batch statistics, and a lagged same-layer operator supplies a covariance signal. Temporal adjacency turns invariance into slowness, making the construction a soft form of Slow Feature Analysis. We show that detaching historical quantities truncates cross-time credit assignment without removing the current-observation gradient. At each layer, a detached regularized target maps its residual to the state of each neuron. The feedforward update then factors into a presynaptic value and this postsynaptic state; one lateral matrix corrects the state once and receives an anti-Hebbian state–state contribution. The resulting encoder processes one observation at a time, sends no error signal across learned layers or time, and stores three causal scalars per neuron together with one matrix of auxiliary lateral parameters per layer. On a class-chunked MNIST stream, batch-size-one stochastic-gradient training reaches $96.39 \pm 0.19\%$ held-out linear-probe accuracy. Random features reach $95.14 \pm 0.27\%$, and a layer-local supervised contrastive network reaches $96.95 \pm 0.13\%$. A spatially and temporally relaxed reference, TEREL-Offline, reaches $98.39 \pm 0.04\%$. Shuffling the stream or removing the temporal term reduces accuracy. Removing variance control produces a nearly constant representation, whereas removing covariance control causes numerical failure. Continued unlabeled updates preserve held-out accuracy when their step size is selected for a trained representation.

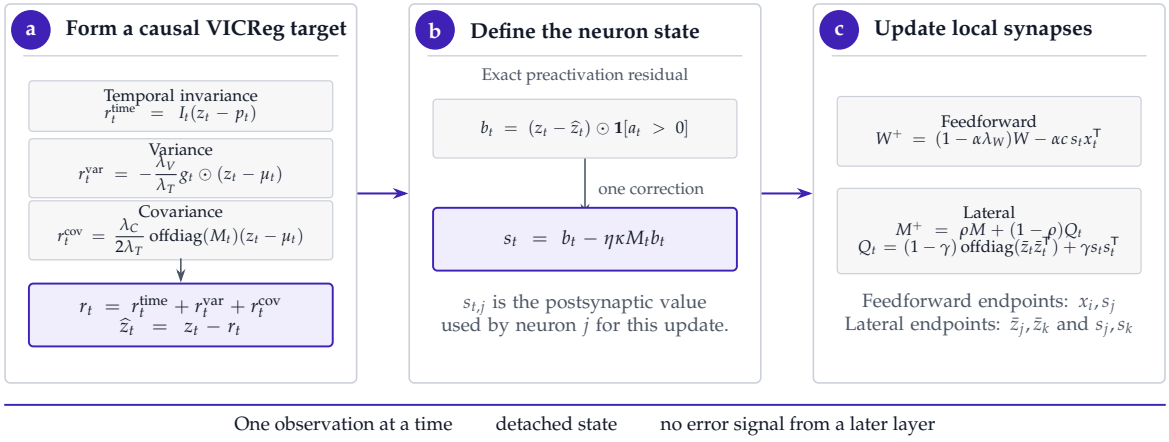


Figure 1: TEREL in one layer. **a**, A causal VICReg surrogate forms the target. **b**, Its residual and one lateral correction define neuron state. **c**, Synapses update from endpoint quantities before output detachment.

1 Introduction

Representation learning needs both a relation that should be preserved and a mechanism that prevents collapse. VICReg separates these roles into invariance between paired views, marginal variance, and cross-coordinate covariance control [1]. If the paired views are consecutive observations, invariance becomes slowness. The resulting combination is a soft form of Slow Feature Analysis (SFA): temporal continuity supplies the learning signal, while variance and decorrelation exclude constant or duplicated coordinates [2]. Temporal-coherence objectives have long used this relation to learn visual invariances [3].

The optimization mechanism matters when observations arrive as a stream. Backpropagation through a sequence retains an activation graph across time; backpropagation through depth also transports a credit signal across learned layers [4]. Predictive-coding and equilibrium formulations can replace that graph with repeated neural dynamics, but then require iterative settling [5, 6]. Both choices complicate continual operation on hardware whose memory, communication, or update schedule is constrained [7, 8]. Here, *local in time* means that the current gradient depends only on the current observation and fixed-size detached quantities, never on a graph through the history that produced them. *Local in space* means that a synapse is updated from quantities available at its two endpoints and within its layer.

Hebbian and anti-Hebbian networks give spatially local learning a useful algebra: similarity-matching objectives yield Hebbian feedforward and anti-Hebbian lateral rules for linear representations [9, 10]. We seek the corresponding nonlinear rule for the invariance–variance–covariance decomposition. Adjacent activations replace paired augmented views, detached running moments replace batch statistics, and a lagged same-layer operator supplies a covariance-gradient proxy. These substitutions define a causal VICReg surrogate, not the original batch loss. They produce a regularized target whose residual combines temporal attraction, variance preservation, and decorrelation. Mapping that residual through the local activation derivative and one same-layer correction gives the signed neuron state used by both feedforward and lateral plasticity.

This construction yields TEREL, a deep nonlinear encoder trained one observation at a time. Adjacent learned layers are separated by a stop-gradient, and all temporal references are detached. The feedforward update is an outer product of presynaptic activity and the postsynaptic neuron state. One lateral matrix serves both target decorrelation and state correction; its state–state component is anti-Hebbian under the inhibitory sign convention. Running moments, lateral parameters, and feedforward weights follow fixed-size leaky recurrences, and the online rule uses plain SGD without optimizer moments.

The experiments compare this local rule with a reference that relaxes both locality constraints. TEREL uses batch size one and two passes over the training observations. TEREL-Offline retains a subsequence, evaluates the soft-SFA loss only at the final layer, and backpropagates through both depth and the subsequence. On MNIST, TEREL improves over random features, a linear input reference, batch SFA, and incremental SFA. It remains within 0.57 percentage points of layer-local supervised contrastive learning despite using no labels in its objective. TEREL-Offline reaches the accuracy of a fully offline supervised network. Matched interventions further show that the temporal order and each regularizer affect the learned representation.

The paper makes three contributions:

1. a causal construction of VICReg's three-part decomposition in which temporal adjacency yields soft SFA, together with a precise account of the current-observation gradient retained under stop-gradient operations;
2. a regularized activation target whose ReLU preactivation residual gives a neuron-local postsynaptic state, producing feedforward gradients with Hebbian two-factor structure, an anti-Hebbian lateral contribution, one explicit same-layer correction, and one auxiliary matrix per layer; and

3. a prespecified comparison with relaxed-locality, supervised, random, and classical SFA references, plus matched mechanism tests.

2 Temporal Regularized Learning

2.1 One observation and a causal VICReg construction

Consider observation t at a layer with input $x_t \in \mathbb{R}^d$ and width D. The layer computes

$$a_t = W_t x_t + q_t, \qquad z_t = \phi(a_t), \tag{1}$$

where $W_t \in \mathbb{R}^{D\times d}$, $q_t \in \mathbb{R}^D$, and z_t is passed to the next learned layer only after detachment. Before this observation, each neuron stores a preceding activation p_t, a running mean μ_t, and a running variance v_t. The layer also stores a lateral matrix $M_t \in \mathbb{R}^{D\times D}$. These quantities are fixed during the current gradient calculation. Let $I_t \in \{0,1\}$ indicate whether p_t belongs to the same sequence, and write $\bar{z}_t = z_t - \mu_t$ and $g_t = [\tau - v_t]_+$ for a variance target $\tau > 0$.

VICReg decomposes its objective schematically as $\mathcal{L}_{\text{VICReg}} = \lambda_I \mathcal{L}_I + \lambda_V \mathcal{L}_V + \lambda_C \mathcal{L}_C$, where the terms enforce invariance between paired views, marginal dispersion, and cross-coordinate decorrelation [1]. We preserve these roles while replacing the second view by p_t, batch moments by the detached μ_t, v_t, and the differentiated batch-covariance signal by the lagged operator $A_t = \text{offdiag}(M_t)$. This gives the samplewise terms

$$\ell_T = \frac{I_t}{D}\|z_t - p_t\|_2^2, \qquad \ell_V = -\frac{1}{D} g_t^\top (\bar{z}_t \odot \bar{z}_t), \qquad \ell_C = \frac{1}{D} \bar{z}_t^\top A_t \,\text{sg}(\bar{z}_t). \tag{2}$$

The layer minimizes $\ell = \lambda_T \ell_T + \lambda_V \ell_V + \lambda_C \ell_C$ with $\lambda_T > 0$. The first term makes consecutive activations similar. The second increases the coordinates whose stored variance lies below τ. The third applies a lagged same-layer decorrelation signal. Detachment makes each term a function of the current activation and bounded stored quantities, without a graph through earlier observations.

This is a causal surrogate of the VICReg decomposition rather than a scalar re-expression of its batch objective. In particular, ℓ_V is a gated instantaneous expansion surrogate, and A_t is a lagged covariance-gradient proxy. We call an update *causal in credit assignment* when its gradient at time t depends only on the current observation, current layer parameters, and fixed-size detached quantities computed from earlier observations. Their retained size may depend on layer width but not on stream length. This is the precise sense in which the rule is local in time.

Proposition 1 (Causal surrogate and current learning signal)**.** *For fixed p_t, μ_t, v_t, M_t, and I_t, the derivative of the weighted surrogate is*

$$\nabla_{z_t} \ell = \frac{2\lambda_T}{D} r_t, \qquad r_t = I_t(z_t - p_t) - \frac{\lambda_V}{\lambda_T} g_t \odot \bar{z}_t + \frac{\lambda_C}{2\lambda_T} A_t \bar{z}_t. \tag{3}$$

It is therefore causal in credit assignment. Before the same-layer state correction, its feedforward gradient is $2\lambda_T (J_{\phi,t}^\top r_t) x_t^\top / D$. Detachment thus preserves a current-observation learning signal whenever $x_t \neq 0$ and $J_{\phi,t}^\top r_t \neq 0$.

Proof. The detached quantities are constants for differentiation. The derivatives of the temporal, variance, and covariance terms are respectively $2\lambda_T I_t(z_t - p_t)/D$, $-2\lambda_V g_t \odot \bar{z}_t / D$, and $\lambda_C A_t \bar{z}_t / D$. Their sum gives Equation (3); the chain rule gives the feedforward outer product. An outer product of two nonzero vectors is nonzero. □

2.2 The regularized target precedes the neuron state

Let r_t^T, r_t^V, r_t^C denote, in order, the temporal, variance, and covariance addends in Equation (3). The three derivatives can then be collected before a synapse is updated:

$$r_t = r_t^T + r_t^V + r_t^C, \qquad \widehat{z}_t = \mathrm{sg}(z_t - r_t). \tag{4}$$

Thus $\widehat{z}_t$ is the activation preferred by the instantaneous balance of slowness, variance, and decorrelation. It is formed within the layer; it is not an external target.

Proposition 2 (Activation-target equivalence)**.** *At the values used to construct Equation* (4)*,*

$$\widetilde{\ell}_z = \frac{\lambda_T}{D}\|z_t - \widehat{z}_t\|_2^2 \tag{5}$$

has the same derivative with respect to z_t as ℓ.

Proof. By Proposition 1, the derivative of ℓ is $2\lambda_T r_t / D$. Since $z_t - \widehat{z}_t = r_t$ at the construction point and the target is detached, Equation (5) has the same derivative. □

The activation residual must next be expressed at the input side of the nonlinearity. Let

$$b_t = J_{\phi,t}^\top r_t, \qquad J_{\phi,t} = \left.\frac{\partial \phi(a)}{\partial a}\right|_{a_t}. \tag{6}$$

For the ReLU encoder used in the experiments, this is the explicit signed quantity

$$b_t = r_t \odot \mathbf{1}[a_t > 0]. \tag{7}$$

No automatic-differentiation graph is needed to obtain it. A single same-layer correction produces the postsynaptic neuron state

$$s_t = b_t - \eta\kappa M_t b_t, \tag{8}$$

where $\eta > 0$ is the correction step and $\kappa \geq 0$ its strength. This is one matrix–vector operation in the feedforward computation. It is not an iterative settling procedure.

Define the detached preactivation target $\widehat{a}_t = \mathrm{sg}(a_t - s_t)$. Its local mean-squared error gives

$$\widetilde{\ell}_a = \frac{\lambda_T}{D}\|a_t - \widehat{a}_t\|_2^2, \qquad \nabla_{W_t}\widetilde{\ell}_a = \frac{2\lambda_T}{D} s_t x_t^\top. \tag{9}$$

Each feedforward synapse therefore receives a presynaptic scalar $x_{t,i}$ and a postsynaptic scalar $s_{t,j}$. This is the two-factor locality of a Hebbian rule; the sign of $s_{t,j}$ determines potentiation or depression under gradient descent. With learning rate α and coupled decay λ_W, the weight recurrence is

$$W_{t+1} = (1 - \alpha\lambda_W)W_t - \alpha\frac{2\lambda_T}{D} s_t x_t^\top. \tag{10}$$

The corresponding bias update uses s_t alone. The output of this layer is detached before it becomes the input to the next learned layer, so no error signal crosses the layer boundary.

2.3 One lateral matrix and bounded causal state

The same matrix supports the covariance signal in Equation (3) and the state correction in Equation (8). After the synaptic update, form

$$Q_t = (1 - \omega)\,\mathrm{offdiag}(\bar{z}_t \bar{z}_t^\top) + \omega s_t s_t^\top, \qquad M_{t+1} = \rho M_t + (1 - \rho)Q_t, \tag{11}$$

with $\omega \in [0, 1]$ and $\rho \in [0, 1)$. The off-diagonal part of M_t enters the regularized target, while the full matrix enters the neuron-state correction. If inhibitory lateral weights are written as $L_t = -M_t$, the

state component of Equation (11) contributes $-(1-\rho)\omega s_t s_t^\top$ to L_t: same-signed neuron states make their mutual lateral connection more inhibitory, which is the anti-Hebbian sign. The representation component preserves the decorrelation signal required by the target.

The three causal scalars of a neuron follow

$$p_{t+1} = z_t, \qquad \mu_{t+1} = \beta\mu_t + (1-\beta)z_t, \qquad v_{t+1} = \beta v_t + (1-\beta)(z_t - \mu_t)^2. \tag{12}$$

At a sequence boundary, only the validity of p_t is cleared; the learned moments and lateral parameters remain. A layer of width D therefore stores $3D$ causal scalars and one boundary-validity bit. It also carries D^2 auxiliary lateral parameters, separate from its feedforward weights. The reported optimizer is plain stochastic gradient descent and adds no optimizer state. For the 784–512–256 MNIST encoder, these categories contain 2,306 causal scalars, 327,680 auxiliary lateral parameters, and 533,248 feedforward parameters.

2.4 The spatially and temporally relaxed reference

TEREL-Offline measures what the same slow-feature principle can learn after both locality constraints are relaxed within a subsequence. A complete subsequence enters a ReLU multilayer perceptron; the soft-SFA objective is evaluated at its final learned layer, and backpropagation differentiates that objective through all preceding layers and observations in the subsequence. No normalization layer is used. Momentum SGD optimizes this reference, so its update is neither neuron-local nor free of optimizer state. It is consequently a performance reference, not an alternative definition of TEREL.

3 Relation to previous work

SFA obtains invariant features by minimizing temporal variation under normalization and decorrelation constraints [2]. Incremental SFA adapts the classical linear construction to streams but retains a whitening stage and linear feature extraction [11]. Other local networks derive linear slow-feature or similarity-matching algorithms with Hebbian feedforward and anti-Hebbian lateral plasticity [9, 10, 12]. TEREL retains this synaptic factorization while making the postsynaptic factor the signed residual of a nonlinear regularized target at each layer.

Modern self-supervised methods also prevent invariant representations from collapsing. Contrastive predictive coding uses temporal context and negative examples [13]; Barlow Twins and VICReg instead regulate variance and cross-coordinate dependence [1, 14]. TEREL gives the VICReg decomposition a causal temporal realization: adjacent observations provide invariance, running moments gate expansion, and a lagged operator approximates the covariance signal. Vectors and matrices that are computed from parameters or activations but held constant with respect to the parameters being optimized are commonly used in training objectives, such as weight-averaged teacher parameters [15], stop-gradient target projections [16, 17], predictions of an earlier model [18], and importance matrices in continual learning [19, 20]. TEREL strongly relies on this design pattern through its leaky variance and covariance estimators.

Layer-local representation learning blocks the backward path between network modules [21–24]; VICReg losses have also been applied independently at successive layers [25]. Placing VICReg losses at separate depths is therefore not itself new. TEREL instead supplies the three signals causally along time and factors them into neuron-local updates. The Layer-local Supervised Contrastive (Layer-local SupCon) comparator trains each layer with a supervised contrastive objective derived from Khosla et al. [26]. It removes cross-layer credit assignment but still uses labels, minibatches, adaptive optimizer state, and batch normalization. By contrast, TEREL constructs its target from temporal and same-layer quantities and processes one observation per update.

Predictive coding and equilibrium propagation replace ordinary backpropagation with local dynamics around an energy or prediction error [5, 6]. Their recurrent settling differs from the single explicit correction in Equation (8). The correction can be evaluated directly in one feedforward schedule, after which both synaptic updates and all state recurrences are complete.

4 Experimental design

The evaluation tests three hypotheses. First, batch-size-one TEREL should learn a useful deep representation despite its locality constraints. Second, the temporal order and each component of the regularized target should affect that representation. Third, continuing the same unsupervised updates along an evaluation stream should preserve the representation when the continuation step is chosen for an already trained encoder.

4.1 Data, selection, and readout

MNIST contains 60,000 training images and 10,000 held-out images [27]. We reserve a stratified 10,000-image subset of the training data for model selection. The remaining images train the encoder and a linear probe. The probe reads the concatenation of both learned layers and is evaluated on the held-out images. All reported central values are means and sample standard deviations over five independently initialized evaluations; paired comparisons use the same initialization indices and 95% Student-t intervals.

The TEREL stream consists of randomly permuted class-homogeneous chunks of length 16. A sequence boundary separates consecutive chunks. Class labels therefore determine the order, as they do in supervised contrastive sampling, but do not enter the target or either synaptic update. The encoder has widths 512 and 256, uses ReLU, and receives two passes over the encoder-training observations: 100,000 batch-size-one updates in total. The selected configuration, probe, split roles, and evaluation indices were fixed from validation results before held-out evaluation. Exact hyperparameters and raw values appear in Appendix A.

4.2 References and interventions

The performance references are chosen to expose distinct relaxations:

- TEREL-Offline uses the same 512–256 architecture and final-layer soft-SFA principle, with subsequences of 128 and end-to-end backpropagation;
- Layer-local SupCon uses labels and a supervised contrastive objective at each learned layer;
- supervised offline uses labels and backpropagation through the complete network, while the linear input reference applies supervised logistic readout directly to the pixels;
- a random 512–256 ReLU encoder receives the same concatenated-layer readout; and
- batch SFA and incremental SFA provide classical linear slow-feature references.

The methods differ in their intrinsic data and optimization requirements; the comparison therefore reports representation quality rather than equal compute. All downstream evaluations use the same split roles and linear-probe procedure.

Four matched interventions examine TEREL itself. One randomly shuffles the stream. Two set the temporal or variance coefficient to zero. The fourth sets the covariance coefficient to zero. Online continuation begins from the same trained parameters and the same fitted probe as fixed-encoder evaluation. Each representation is emitted before the current observation updates the encoder, so an observation can affect only later representations. The continuation update receives no label. A validation-only scale audit selects the largest continuation step that preserves fixed-encoder accuracy and effective rank.

5 Results

5.1 A useful representation from samplewise local updates

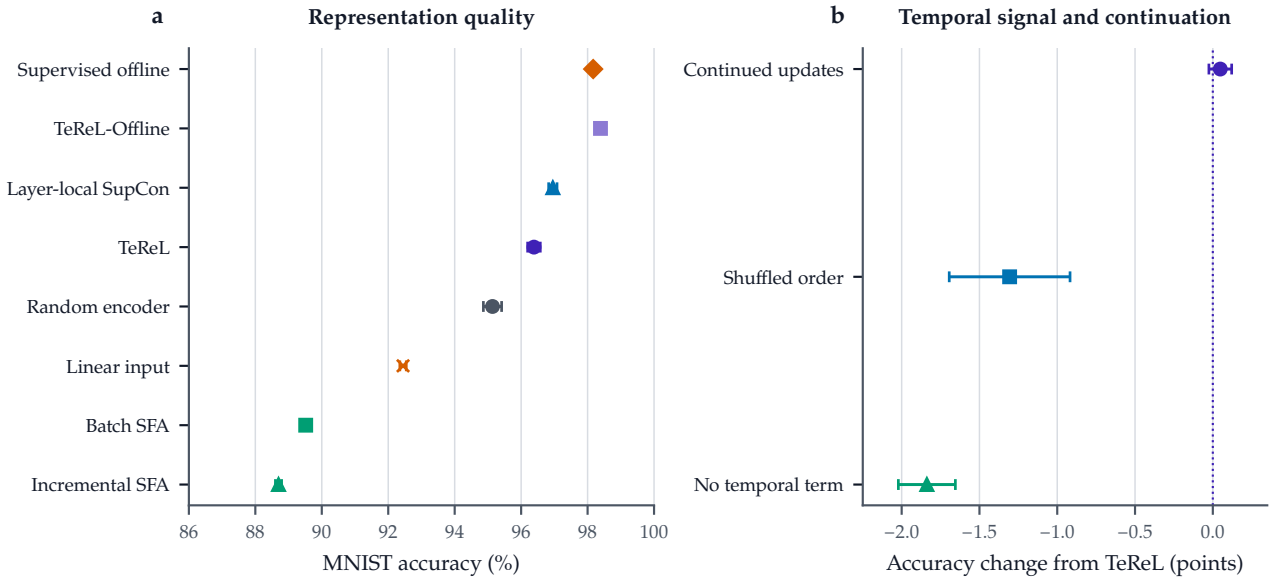


Figure 2: Representation quality and matched temporal effects. a, Held-out linear-probe accuracy. Points show means and horizontal intervals show sample standard deviations. **b**, Paired accuracy changes relative to fixed-encoder TEREL; intervals are 95% Student-t intervals. Online continuation uses the validation-selected step.

TEREL reaches $96.39 \pm 0.19\%$ held-out accuracy (Figure 2a). The random encoder reaches $95.14 \pm 0.27\%$, giving a paired gain of 1.25 percentage points with interval $[0.88, 1.61]$. The gain over a linear probe on the input is 3.94 points $[3.65, 4.23]$. Batch SFA and incremental SFA reach $89.52 \pm 0.13\%$ and $88.70 \pm 0.09\%$, respectively. The nonlinear depth and regularized local target therefore contribute beyond both the input and classical linear slow-feature references.

Layer-local SupCon reaches $96.95 \pm 0.13\%$, 0.57 points above TEREL. This comparator uses class labels, minibatches, batch normalization, and AdamW at every layer, so its 0.57-point advantage comes with a substantially less restrictive learning signal. TEREL-Offline reaches $98.39 \pm 0.04\%$, while the fully supervised offline network reaches $98.17 \pm 0.07\%$. Their proximity shows that the soft-SFA principle itself can support the supervised reference accuracy when subsequence and cross-layer optimization are available. The remaining gap to samplewise TEREL is associated with the locality constraints rather than a weak offline objective.

5.2 Temporal order and anti-collapse terms

The temporal interventions in Figure 2b isolate the contribution of adjacency. Randomly shuffling the stream lowers accuracy by 1.31 paired points, with interval $[0.92, 1.69]$. Removing the temporal term lowers it by 1.84 points $[1.65, 2.02]$. These two changes preserve finite representations but remove the advantage supplied by class-consistent adjacency.

The variance and covariance terms control more severe failure modes. Without the variance term, accuracy falls to $15.10 \pm 0.18\%$, and the median feature variance is approximately 1.4×10^{-5}. The effective-rank statistic is not informative in this limit because many coordinates retain similarly tiny

variance. Without the covariance term, all five evaluations become non-finite before a representation can be measured. The three target components thus play distinct roles: temporal agreement improves the ordered representation, variance prevents a near-constant solution, and covariance stabilizes the multidimensional code.

5.3 Continued online updates require a post-training step size

The encoder and its causal state can continue updating along the evaluation stream without changing the fitted probe. Reusing the initial training step, however, is too aggressive after the representation has formed: a validation-stream diagnostic shows progressively larger losses across stream blocks and a sharp rank reduction. Scaling the continuation step by 10^{-3} preserves validation accuracy (96.17% versus 96.18%) and effective rank (90.41 versus 90.01) while retaining nonzero feedforward parameter change. The same scale is fixed before the held-out continuation comparison in Figure 2b. Across the held-out evaluations, continued updates reach $96.43 \pm 0.15\%$, compared with $96.39 \pm 0.19\%$ for the fixed encoder. The paired change is $+0.05$ points with interval $[-0.03, 0.12]$; effective rank changes by $+0.57$ $[-0.04, 1.18]$. The mean feedforward parameter change is nonzero ($\|\Delta W\|_2 = 0.047$). The rule therefore runs online after training, provided its step size is reduced once the representation has formed.

5.4 Geometry of the learned representation

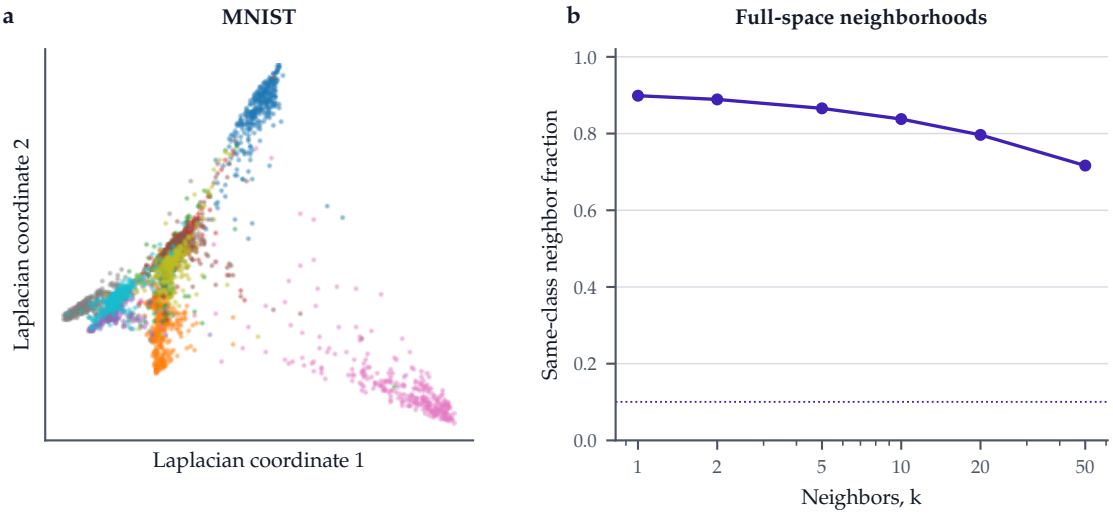


Figure 3: Class structure in the TEREL representation. a, A Laplacian embedding of validation-set concatenated-layer features; colors denote digits and are used only for visualization. **b**, The fraction of same-class neighbors measured in the complete 768-dimensional representation. The dotted line is the class-frequency chance level.

The Laplacian view in Figure 3a shows extended class-specific branches rather than a single collapsed mass. Because any two-dimensional projection can distort neighborhoods, Figure 3b measures the same relation in the complete representation. The nearest neighbor shares the digit label for 89.9% of observations; the fraction remains 79.7% across 20 neighbors and 71.7% across 50, compared with a 10.0% class-frequency baseline. The linear-probe result is therefore accompanied by local class structure distributed through the learned feature space.

5.5 TeReL's neuron state

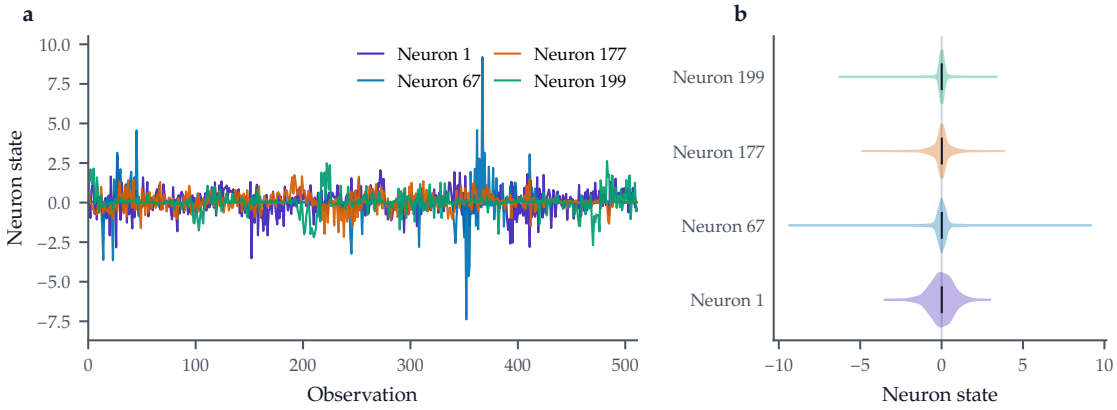


Figure 4: Signed postsynaptic state. a, State trajectories of four randomly selected neurons from the second layer over the same interval. **b**, Their empirical state distributions over the complete validation stream; black bars mark medians.

Figure 4 displays the quantity that multiplies presynaptic activity in Equation (9) during continued samplewise updates on the validation stream. The selected neurons alternate between positive and negative values, so the state supports both directions of synaptic change. Their scales and distributions also differ substantially: some neurons respond through sparse excursions, whereas another carries a broader signed signal. This heterogeneity is hidden by a single state norm but is explicit in the neuron-local formulation.

6 Limitations and future work

The present performance evidence uses a constructed MNIST order. The labels define adjacency but never enter a TEREL target or synaptic update. Natural streams contain several time scales and may require a different balance among temporal, variance, and covariance signals; establishing such a result on large video or sensor streams remains the most important empirical extension.

Continued adaptation also introduces a schedule beyond initial representation learning. A small fixed continuation step preserves the trained code in the current setting, but it does not improve it. Local rules that regulate this step from neuron state, parameter drift, or a stability criterion would make training and long-term inference coincide more fully.

Finally, a layer of width D uses D^2 auxiliary lateral parameters. This is comparable to the feedforward parameters of an equal-width layer and distinct from causal state, but it may still dominate narrow-input or very wide architectures. Sparse, low-rank, or structured lateral recurrences are natural directions for constrained physical implementations. The current experiments also use multilayer perceptrons; convolutional and recurrent architectures would test whether the same target-state construction scales while preserving its locality boundary.

7 Conclusion and outlook

TEREL begins with the invariance–variance–covariance decomposition of VICReg and makes three causal substitutions: adjacent observations for paired views, detached running moments for batch

statistics, and a lagged same-layer operator for the covariance signal. Because the positive relation is temporal adjacency, invariance becomes slowness and the resulting mechanism is soft SFA. The stop-gradients truncate historical credit assignment while preserving the current-observation gradient under the conditions in Proposition 1.

A detached regularized target maps that gradient to a signed postsynaptic state. Presynaptic activity and this state give the feedforward update, while one same-layer matrix supplies an explicit state correction and receives an anti-Hebbian state–state contribution. The complete rule processes one observation at a time with three causal scalars per neuron, one auxiliary matrix per layer, and no error signal across learned layers or time.

The experiments show that this locality is compatible with a useful deep representation. Samplewise TEREL improves on random, input-linear, and classical SFA references; the relaxed TEREL-Offline construction reaches the fully supervised offline result. These properties suit continual representation learning on devices where observations arrive sequentially and global credit assignment is expensive. The derivation also separates the regularized representation mechanism from the relation used to construct positive pairs. SFA is its temporal specialization; other locally available relations could support different causal representation learners. The immediate next steps are natural temporal streams and adaptive long-term updates.

AI Use Statement

The original TEREL framework and initial manuscript predate the generative-AI assistance disclosed here. Generative AI tools were used to assist with revision planning; theoretical and experimental development; implementation and experiment execution; analysis and interpretation; figure creation; literature search and synthesis; and manuscript drafting and editing. The author reviewed AI-assisted code, experimental outputs, numerical claims, citations, figures, and text against saved artifacts and executable code, and takes responsibility for the final content.

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A Evaluation protocol and raw values

Table 1 lists the reported configurations. Model selection uses only the reserved training-data validation split. The configuration, readout, evaluation indices, and repetitions are fixed before the held-out comparison; the held-out values are not used to choose among encoder variants.

Table 1: Encoder configurations used in the reported comparison. A sample is one observation presented to the encoder.

Method	Samples	Batch or subsequence	Optimizer	Normalization
TEREL	100,000	1	SGD	none
TEREL-Offline	1,000,000	128	momentum SGD	none
Layer-local SupCon	3,000,000	256	AdamW	BatchNorm
Supervised offline	500,000	256	AdamW	none
Batch SFA	50,000	complete fit	eigensolver	whitening
Incremental SFA	50,000	1	incremental	whitening

For TEREL, the selected coefficients are $\lambda_T = 1$, $\lambda_V = 3.001981$, $\lambda_C = 0.606133$, $\tau = 1$, $\beta = 0.99$, $\rho = 0.995$, $\kappa = 0.574950$, $\eta = 0.1$, and $\omega = 0.591110$. The learning rate is 0.0138703 with coupled decay 10^{-6}. TEREL-Offline uses 20 epochs, learning rate 0.0109437, momentum 0.95, decay 10^{-6}, variance coefficient 2.529924, covariance coefficient 0.0218471, and variance target 0.862789. The linear probe uses AdamW for 60 epochs with learning rate 0.003, decay 10^{-4}, and batch size 2048.

Table 2: Held-out MNIST accuracy by initialization index. Values are fractions; a dash denotes a non-finite representation.

Method	1101	1202	1303	1404	1505
TEREL	.9625	.9656	.9661	.9618	.9633
TEREL-Offline	.9842	.9838	.9832	.9841	.9841
Random encoder	.9467	.9533	.9513	.9533	.9523
Input linear probe	.9250	.9231	.9254	.9250	.9238
Supervised offline	.9816	.9820	.9823	.9806	.9819
Layer-local SupCon	.9698	.9693	.9714	.9693	.9679
Batch SFA	.8961	.8962	.8954	.8929	.8952
Incremental SFA	.8865	.8883	.8861	.8876	.8864
No temporal term	.9456	.9465	.9455	.9441	.9457
No variance term	.1495	.1493	.1533	.1526	.1504
No covariance term	–	–	–	–	–
Shuffled order	.9508	.9474	.9523	.9512	.9523

B Locality and operation accounting

For a layer of width D with input width d, the feedforward multiplication costs Dd multiply–accumulates. Forming the target and causal statistics is linear in D. The lateral target and state correction each use a $D \times D$ matrix–vector multiplication, and updating the lateral moment uses a rank-one outer product. The persistent quantities are summarized in Table 3; temporary activations are not persistent causal state.

Table 3: Persistent quantities for one TEREL layer of width D and input width d.

Category	Quantity	Size
Causal neuron state	p, μ, v and one layer-validity bit	$3D + 1$
Auxiliary lateral parameters	M	D^2
Feedforward parameters	W, q	$Dd + D$
Optimizer state	none	0